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Searchsmart.org: Guiding researchers to the best databases and search systems for systematic reviews and beyond

Michael Gusenbauer

Institute of Innovation Management, Johannes Kepler University Linz, Linz, Austria

Correspondence
Michael Gusenbauer, Institute of Innovation Management, Johannes Kepler University Linz, Altenberger Straße 69, 4040 Linz, Austria.
Email: michael.gusenbauer@jku.at

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Abstract

When searching for scholarly documents, researchers often stick with the same familiar handful of databases. Yet, just beyond these limited horizons lie dozens of alternatives with which they could search more effectively, whether for quick lookups or thorough searches in systematic reviews or meta-analyses. Searchsmart.org is a free website that guides researchers to particularly suitable search options for their particular disciplines, offering a wide array of resources, including search engines, aggregators, journal platforms, repositories, clinical trials databases, bibliographic databases, and digital libraries. Search Smart currently evaluates the coverage and functionality of more than a hundred leading scholarly databases, including most major multidisciplinary databases and many that are discipline-specific. Search Smart's primary use cases involve database-selection decisions as part of systematic reviews, meta-analyses, or bibliometric analyses. Researchers can use up to 583 criteria to filter and sort recommendations of databases and the interfaces through which they can be accessed for user-friendliness, search rigor, or relevance. With specific pre-defined filter settings, researchers can quickly identify particularly suitable databases for Boolean keyword searching and forward or backward citation searching. Overall, Search Smart's recommendations help researchers to discover knowledge more effectively and efficiently by selecting the more suitable databases for their tasks.

KEYWORDS

bibliometric analyses, comparison of bibliographic databases, database selection, metamorphic testing, systematic reviews and meta-analyses, systematic searching

Highlights

What is already known

- Database selection influences the quality of knowledge discovery, particularly in systematic reviews, meta-analyses and bibliometric analyses.

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- Selecting a few highly suitable literature databases is likely to be enough to fulfill high standards in sample selection.

What is new

- Search Smart is a free website that guides researchers to the literature databases that are likely most productive to search with.
- Based on the original data Search Smart collects, databases' capabilities and limitations become comparable in unprecedented detail.

Potential impact for *Research Synthesis Methods* readers

- With Search Smart, researchers can quickly learn which databases they should use to find relevant documents most effectively and efficiently.

1 | INTRODUCTION

In just 14 years, the number of documents available to researchers has doubled.* Thus, over time, it is becoming increasingly important to know how best to identify relevant scholarly documents such as journal articles, conference publications, and clinical trials. Academic databases accessible through search engines, aggregators, journal platforms, digital libraries, and other search systems are essential sources for researchers to find these documents. To use their research time to best effect, researchers must focus their search on the most suitable databases. This means selecting databases that likely cover the most relevant documents and allow the type of searches that researchers aim to conduct.

However, choosing the most suitable databases can be challenging. There are hundreds of options available, all offering different coverage, functionality, and interfaces. Currently, there is only limited information on which database to select for searching—whether as part of day-to-day work or for more thorough reviews. Database analyses are focused on one or a few databases (e.g., Boeker et al.[1]) or a narrow range of characteristics, such as subject coverage (e.g., Martín-Martín et al.[2]). Most analyses neglect the functionality of search interfaces entirely. This lack of detailed information and rankings means that researchers tend to stay with the databases they already know, or those used by studies they cite, with no clear sense of whether or not these databases are the most relevant or appropriate.

Search Smart (www.searchsmart.org)[3] is a web-based tool to guide researchers to particularly suitable databases to search for academic literature. Search Smart distinguishes between databases that host scholarly documents and the search systems that make the databases accessible via a search interface. Search Smart compares a wide variety of databases and search systems—aggregators such as Web of Science, search engines such as Google Scholar, publisher databases such as SpringerLink, bibliographic databases such as PubMed, digital libraries such as ACM, repositories such as arXiv, or clinical trials databases such as ClinicalTrials.gov. The common denominator is that they all host scholarly bibliographic information of all types.

Currently, Search Smart allows researchers to compare 106 important bibliographic and clinical trials databases (see Table 1), which are searchable via 68 interfaces. This selection is continuously expanding. The databases are selected for analysis when they primarily host scholarly content, cover at least one of the 26 All Science Journal Classification (ASJC) subjects,[4] primarily in English, and vary in size but with a preference for larger databases with over a million records. Additionally, databases must provide a search interface and query hit count information. The selection aims to encompass a diverse mix of open and paywalled, new and established databases, and is ready to include new systems based on these standards.

Search Smart allows independent comparisons at an unprecedented level of detail because it collects original data on databases and their search interfaces unavailable elsewhere. Through detailed comparisons, Search Smart aims to enable users to select suitable databases with just a few clicks—whether that be the most popular choices, options hidden behind paywalls, or unjustly neglected alternatives.

2 | SEARCH SMART'S MAIN USER GROUPS

While Search Smart is designed to assist all academic information seekers, it is particularly valuable for those whose research outcomes are closely tied to the suitability of the databases on which they rely. Within that broad

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TABLE 1 List of the 106 databases and search systems covered as of July 15th, 2024.

1. ABI/Inform Global (via ProQuest)
2. Academia.edu
3. Academic Search Elite (via EBSCOhost)
4. Academic Search Premier (via EBSCOhost)
5. ACM Guide to Computing Literature
6. ACS Publications
7. AIS eLibrary
8. APA PsycInfo (via EBSCOhost)
9. APA PsycInfo (via Ovid)
10. APA PsycNet
11. Arts & Humanities Citation Index (via Web of Science)
12. arXiv
13. Bielefeld Academic Search Engine
14. Biological Science Database (via ProQuest)
15. BIOSIS Citation Index (via Web of Science)
16. Business Source Premier (via EBSCOhost)
17. CAB Abstracts (via Ovid)
18. CAS SciFinder-n
19. CINAHL (via EBSCOhost)
20. CINAHL Plus (via EBSCOhost)
21. ClinicalTrials.gov
22. CNKI Overseas
23. Cochrane Database of Systematic Reviews
24. Cochrane Library—CENTRAL
25. Conference Proceedings Citation Index—Science (via Web of Science)
26. Conference Proceedings Citation Index—Social Science & Humanities (via Web of Science)
27. CORE
28. Crossref
29. dblp
30. Dimensions—Clinical Trials
31. Dimensions—Publications
32. Dimensions—Publications (free)
33. Dissertations & Theses Global (via ProQuest)
34. DOAJ
35. Earth, Atmospheric & Aquatic Science Database (via ProQuest)
36. EconBiz
37. EconLit (via EBSCOhost)
38. EconStor
39. Embase (via Ovid)
40. Emerald Insight
41. Emerging Sources Citation Index (via Web of Science)

(Continues)

TABLE 1 (Continued)

42. Environmental Science Database (via ProQuest)
43. Epistemonikos
44. ERIC
45. ERIC (via EBSCOhost)
46. EU Clinical Trials Register
47. Europe PMC
48. Food Science and Technology Abstracts (via EBSCOhost)
49. GeoRef (via ProQuest)
50. Google Scholar
51. GreenFILE (via EBSCOhost)
52. HAL
53. IEEE Xplore
54. Ingenta Connect
55. International Bibliography of the Social Sciences (via ProQuest)
56. Internet Archive Scholar
57. JSTOR
58. Lens
59. Medline (via EBSCOhost)
60. Medline (via Ovid)
61. Medline (via Web of Science)
62. Mendeley
63. Naver Academic
64. Nursing & Allied Health Database (via ProQuest)
65. Open Access Theses and Dissertations
66. OpenAIRE
67. OpenAlex
68. Overton—Scholarly Articles
69. Overton—Policy Documents
70. Paperity
71. Pascal and Francis Bibliographic Databases
72. Policy Commons
73. Psychology & Behavioral Sciences Collection (via EBSCOhost)
74. Public Health Database (via ProQuest)
75. Public Library of Science (PLOS)
76. PubMed
77. RePEc (via EconPapers)
78. RePEc (via IDEAS)
79. ResearchGate
80. SAGE Journals Online
81. ScanMedicine
82. Science Citation Index Expanded (via Web of Science)
83. ScienceDirect

(Continues)

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TABLE 1 (Continued)

84.	ScienceOpen
85.	Scilit
86.	Scinapse
87.	Scite
88.	SciTech Premium Collection (via ProQuest)
89.	Scopus
90.	Semantic Scholar
91.	Social Sciences Citation Index (via Web of Science)
92.	Social Sciences Premium Collection (via ProQuest)
93.	SocINDEX (via EBSCOhost)
94.	Sociological Abstracts (via ProQuest)
95.	SPORTDiscus (via EBSCOhost)
96.	SpringerLink
97.	SSRN
98.	Taylor and Francis Online
99.	Virtual Health Library
100.	Web of Science Core Collection
101.	WHO International Clinical Trials Registry Platform
102.	Wiley Online Library
103.	World Transit Research
104.	WorldCat—Article/chapter only
105.	WorldCat—Thesis/dissertation
106.	zbMATH Open

category, Search Smart mainly caters to three user groups.

First, systematic reviewers and meta-analysts (from now on, simply "systematic reviewers") need a search process that is comprehensive, reproducible, and transparent.[5] Many disciplines rely on the medical Cochrane Handbook for Systematic Reviews of Interventions[6] and its Methodological Expectations of Cochrane Intervention Reviews (MECIR).[7] These serve as methodological benchmarks[8] across disciplines and inform guidelines such as those from the Campbell Collaboration[9] or the Collaboration for Environmental Evidence.[10] Cochrane mandates *systematic keyword searching* and *backward citation searching*, along with recommended *forward citation searches*. Search Smart allows users to identify particularly suitable options for all three of these search types with pre-defined filter settings. Further, Search Smart provides recommendations for researchers seeking *specific record types*, particularly for medical researchers searching for *clinical trials*.

Second, bibliometric analysts investigate the connections between research documents as tracked by databases' citation indices, whose quality Search Smart compares. The popular VOSviewer[11] software, for example, relies on *backward citation searches* to draw citation links, co-citation links, or bibliographic coupling between documents. Thus, the coverage of the chosen database's backward citation information will determine the quality and shape of the resulting network analyses.[12]

Third, librarians and other information specialists can carry out *detailed comparisons of the databases* they use and curate, obtaining insights into the pros and cons of different databases based on Search Smart's unique data. Search Smart not only gives an overview of search options but also independently verifies search providers' claims.

3 | HOW SEARCH SMART WORKS

Search Smart generates ranked lists of databases that meet specific requirements and that cover more records of a particular type or related to a specific discipline. Search Smart offers 583 criteria with which users can filter and sort databases (see Figure 1, detail A). As these myriad options for filtering and sorting can be overwhelming at first, Search Smart offers three pre-defined filter settings—so-called "presets"—that activate a combination of multiple filters at once (see Figure 1, detail B). Search Smart uses presets to evaluate databases as *extensive*, *sufficient*, or *limited* based on their ability to meet users' requirements—categories that have been developed in previous research.[5] Databases that meet all minimum requirements are deemed *sufficient*, while those that also provide bulk select and export options—capabilities to enable search efficiency and organization of large result sets—are rated *extensive*. Databases that fail one or more of the minimum requirements are labeled *limited* and are recommended to be used with caution or not at all (for details, see https://www.searchsmart.org/faq#what-are-presets; button see Figure 1, detail B).

Search Smart's six primary use cases and their relevance for each user group are outlined in Table 2 and further elaborated below. A case study in Section 4 provides an illustration.

3.1 | Use case 1: Systematic keyword searching

Selecting the **"Systematic keyword searching" preset** activates nine filters that limit the selection of databases to those that support: accurate Boolean OR/AND operators; reproducible search results; verbatim queries

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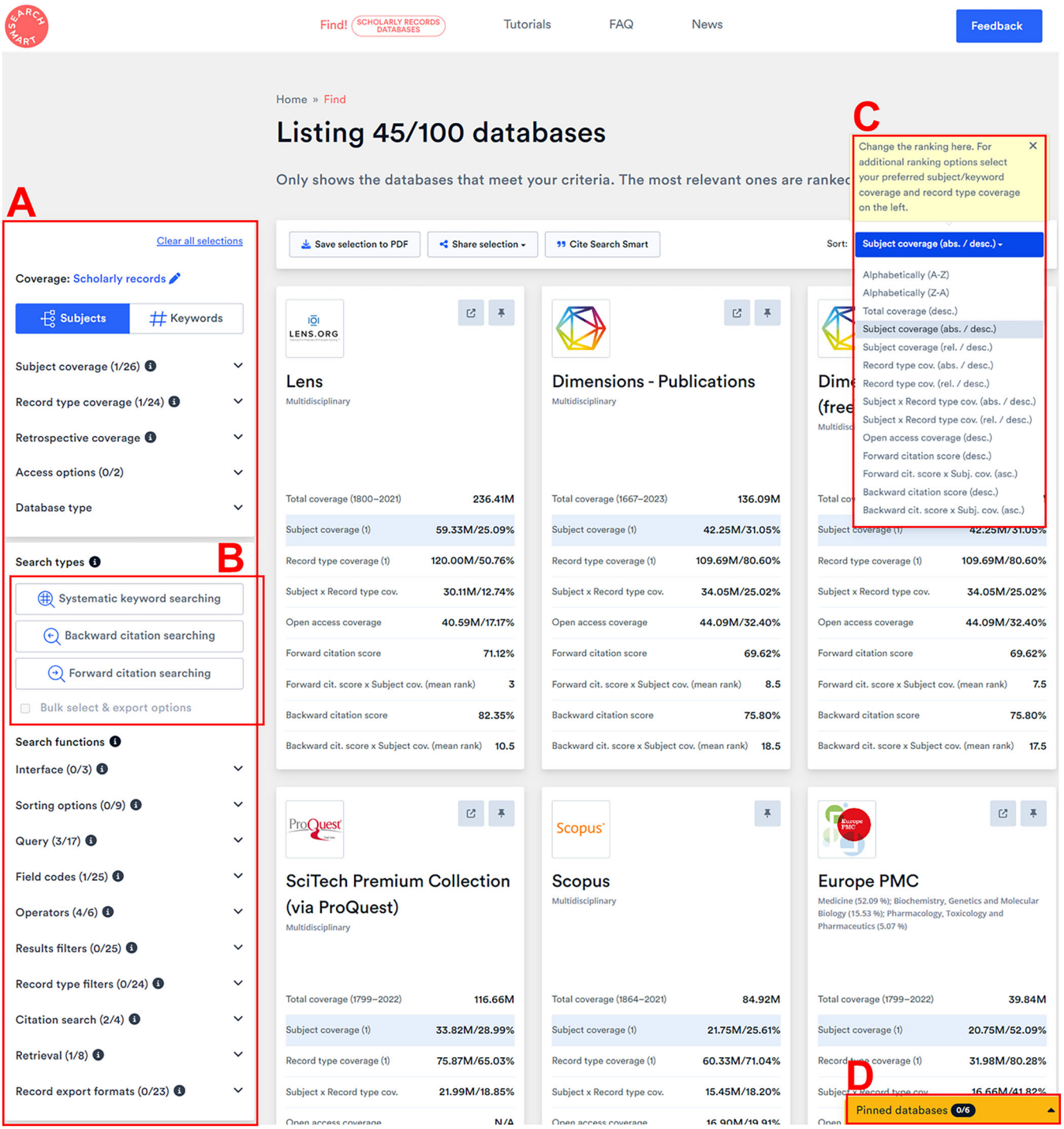

FIGURE 1 Search Smart "Find!" view ranking 45 out of 100 scholarly records databases: Users may filter and sort databases based on their coverages and functions. (A = Coverage and functions filters; B = Pre-defined filters ("presets"), including the "Bulk select & export options" button; C = Sorting options for databases; D = Detailed comparison of up to six pinned databases).

and nested searching; searches with 25 or more keywords; abstract searches; and at least 1000 accessible search results. These are the minimum standards that databases and their search systems must fulfill if systematic reviewers are to rely on them as principal databases.[5]

Additionally selecting the **"Bulk select & export options" button** will activate two more filters that further limit the selection by requiring databases to support bulk selection of records and exports of 50 or more records at a time. Further, systematic reviewers may select additional filters to list only databases that support

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TABLE 2 Search Smart's main user groups and their use cases.

Search Smart use case	Systematic reviewers, meta-analysists	Bibliometric analysts	Librarians, information specialists
(1) Systematic keyword searching	•		•
(2) Backward citation searching	•	•	•
(3) Forward citation searching	•		•
(4) Searching for specific record types	•		•
(5) Clinical trials searching	• (mostly health sciences)		• (mostly health sciences)
(6) Detailed database comparisons			•

functions such as truncation, the Boolean NOT operator, a search history, a query builder, proximity operators, or exceptionally long queries. Experimenting with activating/deactivating filter and preset settings is straightforward, allowing users to learn which databases support which functions.

Systematic reviewers are recommended to search the potentially more fruitful databases first,[13] likely a combination of a few, yet highly suitable multidisciplinary and specialized databases.[14] Thus, users will be interested in databases that cover most of the records they seek. In addition to filtering databases based on their systems' functions ("Systematic keyword searching" preset), Search Smart allows databases to be sorted based on their coverage (see Figure 1, detail C).

To rank databases *with the most records* from a particular discipline at the top, users may select one or more subjects and sort according to absolute coverage via **"Subject coverage (absolute/descending)."** To identify the most specialized databases, users may sort via **"Subject coverage (relative/descending)"** which will rank databases *with the highest share* from a particular discipline at the top. High *absolute* coverage supports sensitive searches (finding more relevant records), while high *relative* coverage supports precise searches (finding fewer irrelevant records). Similarly, databases may be also sorted according to their record-type coverage, although this information is not available for all databases. If authors seek specific record types (e.g., journal articles, conference proceedings), they can combine Search Smart's subject and record-type coverage filters to show the databases covering the most records of particular types on specific subjects.

3.2 | Use case 2: Backward citation searching

Backward citation searching is the search for references cited in a specific seed document. Since some databases do not support this function, researchers must select the **"Backward citation searching" preset**, which activates two filters: backward citation information available and 1000 or more accessible records. Together, these two criteria ensure that the database's interface shows backward citation information and that sufficient records will be displayed. Many users will want to efficiently export backward citations for subsequent deduplication and organization in reference management software. For this purpose, selecting the **"Bulk select & export options" button** will add two more filters: bulk select multiple records' backward citations and bulk export 500 or more backward citations at a time. If users have further requirements, they may add additional filters to identify databases that meet them.

Users will probably choose to sort the filtered databases by "**Backward citation score (descending)**": an indicator of the comprehensiveness and accuracy of a database's backward citation index. Citation scores vary between 0% (no backward citation information) and 100% (all records have fully accurate backward citation information). This sorting option is only available when the "backward citation information" filter is active. Further, since ranking solely on citation score will place databases highly irrespective of their subject focus, users can use a combined filter—**"Backward citation coverage × Subject coverage (descending)"**—to identify those databases that offer greater absolute coverage of one or more selected subjects combined with a higher backward citation score.

Ranking according to backward citation score helps to identify databases that likely provide more comprehensive and accurate data upon which bibliographic analysts can base their analysis. As the popular VOSviewer,[11] Bibliometrix,[12] CiteSpace[15] software programs can only compute data from a single citation index, selecting the most appropriate dataset is imperative. Selecting the **"Supported by bibliometric tools" button** allows filtering for databases that are supported by VOSviewer, Bibliometrix, or CiteSpace. Applying the **"Backward**

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citation coverage × Subject coverage (descending)" sorting to various subject selections shows that, among the aforementioned databases, Lens and Web of Science (various databases) often rank highest. However, individual user requirements may make other databases more suitable. Overall, selecting a database with higher backward citation searching quality will yield more relevant documents.

3.3 | Use case 3: Forward citation searching

Forward citation searching is the search for documents that cite a specific document. The **"Forward citation searching" preset** and the **"Bulk select & export options" button** activate filters similar to those for backward citation searching.

If databases are sorted according to their "**Forward citation score (descending),"** Search Smart will list databases with more comprehensive forward citation indices at the top. This option will only be available when the "forward citation information" filter is active. With the **"Forward citation coverage × Subject coverage (descending)"** setting activated, databases with higher absolute coverage of the selected subject(s) combined with a higher citation score will rank toward the top.

3.4 | Use case 4: Searching for specific record types

Often, users seek specific record types such as journal articles, reviews, preprints, or gray literature. Here, databases vary widely in their coverage: some focus exclusively on specific record types, such as preprint or thesis repositories, while others offer a greater variety.

At Search Smart, users may filter for databases covering specific record types by selecting one or more types in the **"Record type coverage"** filter. Further, using the **"Record type coverage (absolute or relative/descending)" sorting** will rank the databases with the most absolute or relative coverage of the selected record type(s) at the top. If users are looking for records on specific subjects, they can sort according to the combination of subject and record type coverage: **"Subject × Record type coverage (absolute/relative − descending)."** The resulting list will show databases likely containing the most records of a particular type(s) on a specific subject. Here, it is important to bear in mind that some databases, like Google Scholar, are ranked at the very bottom because although they may cover a given type of record, the *extent* to which they cover it is unknown.

3.5 | Use case 5: Clinical trials databases

For systematic reviewers in the medical field, it is mandatory to search for clinical trials according to Cochrane's MECIR guidance.[7] For this use case, the **clinical trials view** of Search Smart solely lists databases that cover clinical trials. Some of these databases will cover clinical trials exclusively (e.g., ClinicalTrials.gov), while others have only a minor share. In the clinical trials view, users may refine clinical trials databases with the same filters as scholarly records databases—particularly via the systematic keyword searching and forward/backward citation searching presets.

3.6 | Use case 6: Detailed database comparisons

Librarians and other information specialists who regularly work with databases may find Search Smart's **detailed database comparisons** useful. In the "Find!" view, users can select ("pin") two to six databases they want to compare in more detail (see Figure 1, detail D). Librarians may use the original data collected at Search Smart to inform procurement decisions by comparing details of proprietary databases against free alternatives, databases offered by different search providers, or different database versions. Librarians may be surprised to learn how almost identical databases differ in subject coverage, retrospective coverage, or search interface functionality (see Figures 2 and 3). Search Smart's original data illustrates performance differences[5] or previously undocumented gaps in some databases' retrospective coverage.

3.7 | Additional use cases

An overview of all six aforementioned use cases and others, along with step-by-step instructions on how to navigate Search Smart, can be found at Searchsmart.org/tutorials. Additional use cases covered include identifying databases with the most coverage of a keyword, for example, "Covid-19"; finding free-to-access databases or those with substantial open access coverage; finding databases with certain retrospective coverage or that support the identification of "hot" papers; databases that support full-text searching, controlled vocabularies, certain field codes, or search operators; and many other search options.

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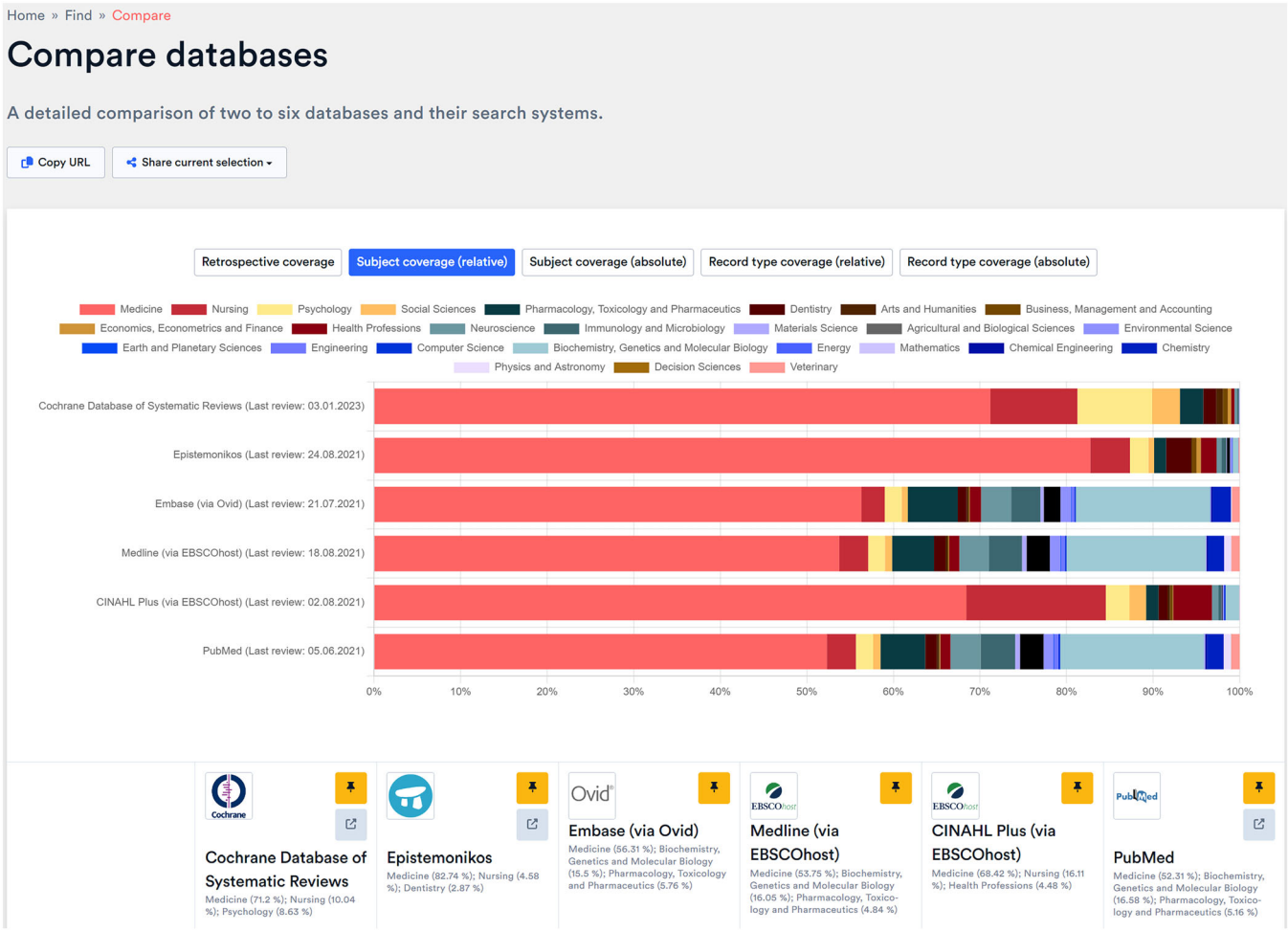

FIGURE 2 Search Smart "Compare" view: A detailed comparison of two to six databases and their search systems. Compares retrospective coverage, subject coverage (absolute/relative), record type coverage (absolute/relative), and many functionality criteria in great detail. This example compares medical databases.

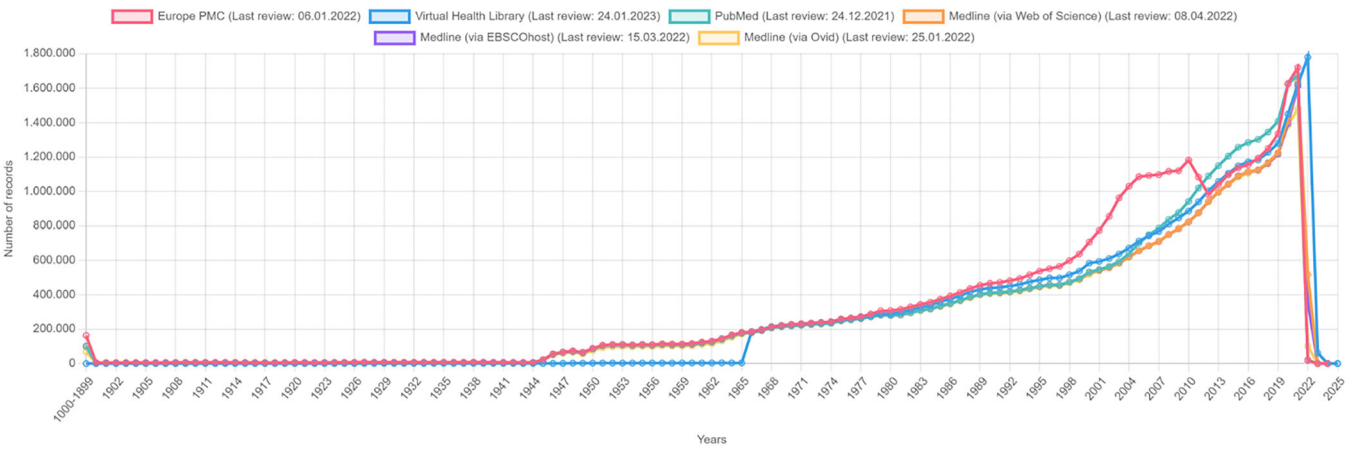

FIGURE 3 Search Smart "Compare" view. A comparison of the retrospective coverage of PubMed and Europe PMC shows the years where each database has unique coverage.

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3.8 | Saving Search Smart's recommendations

A specific advantage of Search Smart is that all selected and sorted databases, including the particular settings used by researchers, may be saved through a PDF export (see the example in Figure 4). The PDF export also tracks which databases were excluded and the filters used to exclude them. Detailed data on individual databases can also be downloaded via Excel exports. Such downloads are recommended to save a snapshot of the databases' functionalities when they were last reviewed. Both PDF and Excel exports contain links so that the recommendations may be updated in the future. Additionally, a share function allows users to save and share specific selections, database details, comparisons, and tutorials with colleagues.

4 | CASE STUDY: USING SEARCH SMART IN A MEDICAL SYSTEMATIC REVIEW

The following illustrates multiple Search Smart use cases in the case of a medical systematic literature review. The medical field was selected because most systematic reviews are conducted in the discipline,[16] making it particularly crucial to identify capable medical databases and search systems. However, this specific illustration also shows how scholars in other areas can identify suitable databases and search systems, as the logics of database selection will be broadly similar.

4.1 | Systematic keyword searching

Medical guidance stipulates systematic keyword searching as the *main* way to identify literature in systematic reviews[6]—a recommendation that is common among disciplines.[9,10] Methodical papers show that searching multiple highly suitable *principal* databases yields best results[17,18]—ideally a combination of a few large, multidisciplinary databases with smaller ones with subject-specific specialization.[19–21]

To identify principal databases that particularly suitable for systematic keyword searching, we select Search Smart's **"Systematic keyword searching" preset** and obtain a filtered list. To further refine this list and identify the largest databases with the widest medical coverage, we select "Medicine" as a subject, sort the list via **"Subject coverage (absolute/descending)"** and get: (1) Lens, (2) Dimensions, (3) SciTech Premium Collection (via ProQuest), (4) Scopus, (5) EuropePMC, (6) Embase (via Ovid), and (7) Web of Science Core Collection. To identify the databases that are most specialized in medicine, we sort via **"Subject coverage (relative/descending)"** and get: (1) Epistemonikos, (2) Cochrane Database of Systematic Reviews, (3) CINAHL Plus (via EBSCOhost), (4) Embase (via Ovid), (5) Virtual Health Library, and (6) Nursing & Allied Health Database (via ProQuest).

Based on insights such as these, systematic reviewers can reflect on which database combination makes the most sense for them. We select the two multidisciplinary databases with the highest *absolute* subject coverage (**Lens and Dimensions**) and one or two databases with the highest *relative* coverage for each subject the review covers (**Epistemonikos and Cochrane Database of Systematic Reviews**). Since the two most medically specialized databases in our ranking are relatively small, we consider the next two to be our principal databases: **CINAHL Plus (via EBSCOhost) and Embase (via Ovid)**.

Medical researchers may have specific additional requirements, and thus may further refine the list with additional filters before firming up on their selections. For example, if they only want databases with extensive export options, they may select the **"Bulk select & export options" button**, or if they want to filter for databases with a hierarchical controlled vocabulary, Search Smart has a filter to suit: **"Query > Controlled vocabulary – hierarchical."** Researchers from resource-constrained environments[22] who cannot access proprietary databases may identify suitable options by additionally activating the **"Non-paywalled databases"** filter.

4.2 | Backward citation searching

Using backward/forward citation searching will improve the quality of a search, especially for linguistically diverse topics that are difficult to capture by keyword searches alone.[23] To identify ideal medical databases for backward citation searching, we select the **"Backward citation searching" preset** and sort the list via **"Backward citation score × Subject coverage (descending)"** based on the Medicine subject we have already selected.

To effectively download citations for subsequent deduplication and screening, we need highly capable databases with extensive export options. Thus, we additionally select the **"Bulk select & export options" button** and receive (1) Lens, (2) Web of Science Core Collection, (3) Scopus, (4) Science Citation Index Expanded (via Web of Science), (4) Medline (via Web of Science), (5) Social Sciences Citation Index (via Web

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Search Smart Report
Databases meeting your criteria

Retrieved on 15.07.2024 11:51
Update report with current data

The criteria you selected:

Subject coverage: Medicine | Record type coverage: Journal publications | Field codes: Abstract | Longest search string (narrow field code): ≥25 | Boolean OR | Verbatim searching | Boolean AND | Boolean work exact | Nested search (parentheses) | Number of accessible records: ≥1000 | Backward citation information | Forward citation information | Sorted by: Subject coverage (abs. / desc.)

Your criteria are met by **45 out of 100 databases:**

Search system - Database	Abs. Subject Cov.	Open details	Search database
Lens	59.327.457	Open details	Search database
Dimensions - Publications (free)	42.251.948	Open details	Search database
Dimensions - Publications	42.251.948	Open details	Search database
SciTech Premium Collection (via ProQuest)	33.816.408	Open details	Search database
Scopus	21.750.619	Open details	-
Europe PMC	20.752.265	Open details	Search database
Web of Science Core Collection	20.165.794	Open details	-
PubMed	18.413.927	Open details	Search database
Medline (via Web of Science)	18.262.833	Open details	-
Science Citation Index Expanded (via Web of Science)	18.127.603	Open details	Search database

FIGURE 4 PDF export to save Search Smart's database comparison. In this example, databases listed toward the top host the most medical records and meet systematic keyword searching criteria.

of Science). The top-ranked databases cover more medical records and have higher coverage and accuracy of backward citations. The top database is **Lens**—a capable choice, but the quality of its backward citation information is somewhat flawed (score of 82.35%). The best option, with a superior citation score yet somewhat inferior subject coverage, is therefore the **Web of Science Core Collection** (score of 94.17%). However, it is important to bear in mind that the Web of Science Core Collection contains different database versions and should only be considered if a researcher has access to its main databases (e.g., SSCI, SCI) and they offer sufficient retrospective coverage.

4.3 | Forward citation searching

The **"Forward citation searching"** preset with the sorting option **"Forward citation score (descending)"** lists databases with the most forward citation coverages at the top. To tailor results to medicine, we also select the Medicine subject and the sorting option **"Forward citation score × Subject coverage (descending)."** To filter out databases with inadequate **"Bulk select & export options,"** we also activate this filter. The resulting list ranks databases with higher medical coverage and more comprehensive forward citation indices toward the top; they are: (1) Lens, (2) Scilit, (3) Scopus, (4) CAS SciFinder-n, and (5) Medline (via Web of Science).

Google Scholar—the gold standard in forward citation coverage—is not included in the list as it does not natively support bulk exports. A workaround may be the Publish or Perish software,[24] which allows downloading of up to 1000 forward citations per publication year for individual seed documents. If researchers seek particular types or qualities of citations, they may access more limited and specialized databases—such as Scopus, which indexes higher-quality journals, or Medline (via Web of Science), which indexes mostly medical documents.

4.4 | Searching for specific record types

Systematic reviewers must include various record types in their analysis—particularly journal articles, conference proceedings, previous literature reviews, gray

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literature, and preprints. Different databases cover these record types to varying degrees. Thus, to allow targeted searches, Search Smart tells reviewers which databases offer the most coverage for these types. Medical reviewers seeking gray literature, for example, select Medicine as the preferred subject, select the **"Gray literature" filter** under record type coverage, and sort the databases via the **"Subject × Record type coverage (absolute/descending)"** option. The resulting list reveals that (1) Bielefeld Academic Search Engine, (2) Lens, and (3) Scite cover most medical gray literature among the databases analyzed at Search Smart. In the case of medical preprints, this list would be: (1) Dimensions, (2) Scilit, and (3) OpenAIRE. If medical researchers are interested in databases that specialize most in medical reviews, they may select the "review" filter under record type coverage and the **"Subject × Record type coverage (relative/descending)"** sorting. In this case, the resulting list is: (1) Cochrane Database of Systematic Reviews, (2) Epistemonikos, and (3) SAGE Journals Online. (Note that record type information is currently unavailable for Core, Google Scholar, Mendeley, Naver Academic, and Semantic Scholar. Thus, even though these databases may indeed cover a specific record type, they will still be ranked at the bottom).

4.5 | Clinical trials databases

When relevant to the topic, medical systematic reviewers must search clinical trials databases.[6,7] In the **Clinical trials view**, Search Smart lists the largest clinical trials databases. By default, Search Smart sorts the most comprehensive trials databases at the top, with Cochrane Library—CENTRAL currently being the most comprehensive. If researchers need to employ specific searches, such as systematic keyword or forward or backward citation searches, they can simply select the appropriate preset to further refine the list (see respective use cases).

4.6 | Resulting selection of databases for a medical systematic review

Based on the previous selections, Search Smart recommends medical systematic reviewers to start with these principal databases as a minimum (any supplementary databases may be added on top): Lens, Dimensions, CINAHL Plus (via EBSCOhost), and Embase (via Ovid) for *systematic keyword searches*, Lens or the Web of Science Core Collection for *backward citation searches*, Lens or Google Scholar for *forward citation searches*, Bielefeld Academic Search Engine as the database with most medical *gray literature*, and Cochrane Library—CENTRAL with most *clinical trials*. These recommendations reflect guidance on combining different search techniques[6] and include principal databases that are functionally more suitable and offer superior relative/absolute coverages compared to the four recommended in a popular study of optimal database combinations, for example.[19]

As researchers may have different coverage and functionality requirements, these suggestions can easily be refined by adding or adapting filter settings. More specifically, when systematic reviewers cannot access the most highly recommended databases, they can quickly determine their next best alternatives. The assistance of librarians and other information specialists may help to specify search strategies further.[25]

5 | HOW SEARCH SMART WAS DEVELOPED

Search Smart was developed with funding from the Austrian Science Fund to create a user-friendly tool that allows users to quickly assess which databases are most suitable for their search needs.[26] Search Smart builds on peer-reviewed research,[5,27,28] introducing and refining various scientometric methods to collect diverse database information (for an overview, see Supplementary Material, Data S1). These methods permit databases' coverage and functionality to be interrogated without system access so Search Smart can independently verify or supplement information published by database providers or university libraries. This is crucial, as search providers rarely describe search options—or, worse, a surprisingly high proportion of information available is inaccurate. For example, metamorphic testing detected unreported issues with search string interpretation in every fourth database.[5] Without such tests, these issues would have remained undetected, search providers would keep on advertising this function, and researchers would continue to rely on it. For many of the 583 criteria on which Search Smart collects information, it is also the only information provider.

To provide accurate coverage and functionality information, Search Smart relies on five unique data collection methods: Query Hit Counts (QHC),[28] backward and forward citation scores,[29] Basket of Keywords (BOK),[27] metamorphic testing, and interface reviews.[5] All these methods have the advantage that they can independently collect information without the support of database providers. Using these methods supports Search Smart's goal to collect data from all different types of academic literature databases—that is, aggregators, journal platforms, repositories, search engines, clinical trials databases, bibliographic databases, and digital libraries.

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5.1 | Limitations

Search Smart's innovative methods enable the unprecedented comparison of coverage and functionality across a wide array of databases. However, its insights have some inherent limitations. First, while estimates of subject coverage are good, and often the only ones available, they are still estimates. Thus, Search Smart's rankings should be seen not as 100% accurate, but rather as informative about the orders of magnitude that a database is likely more or less suitable compared to others (details are discussed here[27]). Coverage estimates are likely less representative for ASJC subject categories covering diverse sub-fields such as the social sciences. Further, Search Smart's 26-category ASJC system does not allow filtering for sub-disciplines. This is less of an issue for disciplines such as medicine or psychology, which are well represented by the classification system. However, researchers from a social science sub-discipline such as archaeology or criminology may find it harder to identify suitable databases based on the less granular 26-category ASJC system. These researchers may resort to their respective super-category, yet Search Smart is likely more suitable to assess larger, multidisciplinary databases and less suitable for smaller specialized ones. Moreover, while the database coverage of Search Smart is increasing, specialized smaller databases—particularly those with a focus on a sub-discipline—are not exhaustively represented. Future updates may bring a more granular classification structure, solving some of these issues.

Second, while Search Smart displays one of the largest selections of large scholarly databases covering mainly English-language documents, it does not cover *all* scholarly databases by any means. This focus is necessary in order to achieve comparability through Search Smart's focused data collection methods. While many of the databases on Search Smart also cover some non-English literature, users specifically seeking non-English or specific regional information will need to look elsewhere.

Third, ideally, we would know the *overlap* of all potentially relevant databases to select a combination with broad joint coverage—that is, the combination that exhibits the least overlap. Unfortunately, Search Smart's data collection methods do not allow this. To validate and supplement its recommendations, users may read multidisciplinary overlap comparisons—however, they are limited to a few databases[2,30,31]: Google Scholar, Scopus, Web of Science Core Collection, Dimensions, Crossref, and Microsoft Academic.

Fourth, search systems tend to innovate only after prolonged periods of stasis. Google Scholar, for example, has barely improved its functionality in recent years. While Search Smart regularly updates its data to reflect databases' functionality based on automatic tests and manual reviews, it will occasionally include outdated assessments. Search Smart encourages users to let it know if they spot errors or outdated information, which Search Smart will aim to resolve as soon as possible. To inform users about the timeliness of the data underlying its recommendations, Search Smart reports data retrieval dates for all tests and reviews in the database details and its Excel exports.

To achieve the best outcomes, researchers should keep all these considerations in mind. Then, with the help of Search Smart, studies that rely on suboptimal or entirely inadequate databases[32] could improve their quality.

6 | CONCLUSION

Overall, we believe Search Smart's recommendations are a significant step toward scholars finding and using the most suitable databases for their research. Search Smart currently covers databases that collectively receive over 1 billion visits per month[†] and are a cornerstone of researchers' work routines worldwide. While most researchers search with the same familiar handful of databases, Search Smart guides them to discover superior alternatives, making their search routines more productive. Search Smart users can learn to search with confidence among the most important English-focused academic databases—both innovative newcomers and established stalwarts, whether free or less accessible. Search Smart highlights that all databases are different, and should only be used when they fit the purpose at hand, that is, when search heuristics—such as keyword or citation searching—match their capabilities (see a recent commentary[26]).

Search Smart encourages researchers to demand more from search systems and consider switching to better options. Increased switching behavior, in turn, should fuel healthy competition among search providers, encouraging them to improve functionality essential to researchers in their pursuit of good science. As new search systems appear each year and incumbents periodically change, Search Smart aims to become a resource that researchers can regularly check to see how their go-to systems compare with innovative newcomers. It is our aspiration that database providers enhance their services based on Search Smart's insights—a symbiotic relationship that is sure to improve the discovery of knowledge.

AUTHOR CONTRIBUTIONS

Michael Gusenbauer: Conceptualization; investigation; funding acquisition; writing – original draft;

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writing – review and editing; visualization; validation; methodology; software; formal analysis; project administration; resources; supervision; data curation.

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CONFLICT OF INTEREST STATEMENT

The author declares no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

ORCID

Michael Gusenbauer https://orcid.org/0000-0001-7768-2351

ENDNOTES

* According to Scopus data: the database hosted about 86 million records in 2022, whereof 44 million had been published since 2009. Similarly, 2.2 million records were published in 2009; by 2022 this had risen to about 4 million.

† At present, Search Smart analyzes 106 paywalled and non-paywalled databases (of which six exclusively cover clinical trials). Non-paywalled systems analyzed at Search Smart are accessed about a billion times per month. Together with paywalled systems' total page visits, the total will likely considerably exceed 1 billion/month. Calculations are based on data from SimilarWeb, a major provider of website analytics. The analysis excludes the paywalled databases of SCOPUS, Web of Science, ProQuest, EBSCOhost, Ovid, and CAS SciFinder-n, for which no reliable traffic data are available. Data are collected for users navigating a specific sub-domain. Most of the domains are search-focused (e.g., scholar.google.com). Yet, some (sub-)domains also involve other services such as publishing (e.g., sciencedirect.com). Data for Google Scholar's 75 top-level domains were aggregated. The data show that a few databases receive disproportionally high traffic, while most are rarely used—despite being highly capable.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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